



RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

SDDC Manager Host-Inventory Desync & Sync Inventory (Brownfield)

Reconciling ESXi hosts stuck unassigned (status:None) in SDDC Manager after an out-of-band vCenter redeploy — the Sync Inventory path, its 504/collision traps, kill-the-hung-process recovery, and the confirmed self-service resolution.

v1.1

VMware Cloud Foundation 9

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INTERNAL

SDDC Manager Host-Inventory Desync & Sync Inventory (Brownfield)

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Platform:** VMware Cloud Foundation 9.0.1 (brownfield-imported management domain)

When to use. SDDC Manager / VCF Operations shows one or more ESXi hosts as **unassigned** / **status: None** with no cluster or domain reference — even though the hosts are **live, connected members** of the vCenter cluster. Typically surfaces after an **out-of-band vCenter redeploy/restore**. This runbook covers the diagnosis, the Broadcom-prescribed **Sync Inventory** reconciliation, its failure modes on a degraded appliance, and the self-service recovery path (**no Broadcom support required**).

1. Symptoms

- **VCF Operations → Fleet Management → Passwords → "Show ESX Hosts"** shows **no ESXi hosts** (the credential inventory has none).
- A component count that **omits ESX** (e.g. "3 Total Components: NSX, vCenter, SDDC" for a 4-host domain).
- Lifecycle / host operations behave oddly for the affected hosts.

2. Diagnosis (read-only, via SDDC API)

```
# Token
TOK=$(curl -sk -X POST https://sddc-manager.lab.local/v1/tokens -H "Content-Type: application/json" \
-d '{"username":"administrator@vsphere.local","password":"<sso-pw>"}' | python3 -c "import
sys,json;print(json.load(sys.stdin)['accessToken'])")

# Host records – look for status=None / no cluster/domain ref
curl -sk -H "Authorization: Bearer $TOK" https://sddc-manager.lab.local/v1/hosts | python3 -c "import
sys,json;[print(h['fqdn'],h.get('status'),(h.get('cluster') or {}).get('id','-')) for h in
json.load(sys.stdin)['elements']]"

# Cluster membership – the cluster usually still lists ALL hosts
curl -sk -H "Authorization: Bearer $TOK" https://sddc-manager.lab.local/v1/clusters | python3 -m json.tool

# Credentials – a brownfield gap shows only svc-vcf-esxiNN SERVICE accounts, NO ESXi root
curl -sk -H "Authorization: Bearer $TOK" "https://sddc-manager.lab.local/v1/credentials?resourceType=ESXI"
| python3 -m json.tool
```

The tell: `/v1/clusters` lists all hosts, but `/v1/hosts` shows some with `status=None` and no back-reference → a **one-directional inventory desync** (cluster→host intact, host→cluster/domain broken). Confirmed 2026-07-27: esxi01 ASSIGNED , esxi02/03/04 status=None ; mgmt domain 95727d6d-... , cluster dc5f4a5b-...

3. Root Cause

Fallout from an **out-of-band vCenter redeploy** on a **brownfield-imported** VCF. The cluster was rebuilt (so it lists all hosts) but the individual host objects never regained their status/cluster/domain references, and ESXi root was never registered into the VCF credential store.

4. The Prescribed Fix — Sync Inventory (Configuration Drift)

Broadcom's documented reconciliation for out-of-band vCenter changes. **Do it in the VCF Operations console** (SDDC Manager UI is deprecated):

Inventory → **Detailed View** → **expand VCF Instances** → **click the domain** → **Actions** → **Sync Inventory** → **confirm**.

Under the hood this runs the **VCF Import Tool**: `vcf_brownfield.py sync --domain-name <domain> ...`.

Two settings that matter in the confirm dialog: - Leave **"Enable Edge cluster sync"** **UNCHECKED** unless you intend it — checking it does a **one-time reset of NSX Edge node passwords** (unnecessary scope/risk for a host-inventory fix). - Check **"I acknowledge..."** (thumbprints) — benign; required to proceed (trusts your own Operations nodes' certs).

5. The Failure Modes (verified 2026-07-27) — and how to avoid them

Trap 1 — the 504 that isn't a failure. The UI call to the sync plugin (`.../plugin/api/v1/domains/<id>/synchronizations`) returns **504 gateway timeout** because the sync takes longer than the HTTP timeout (the dialog even says "up to 3 minutes"). **The backend sync keeps running** (`Sync domain mgmt tasks go In Progress`). **Do NOT re-click on the 504.**

Trap 2 — the N-way collision. Each 504-driven retry launches **another** concurrent sync. Concurrent runs step on each other's assessment/datasource state and fail with: `IMPORT_OPERATION_FAILED · Result status code from set datasource: 400 · Assessment results not found for run with id ... · Failed to sync domain mgmt`. Some also **hang In Progress** indefinitely, holding domain locks.

Rule: click Sync Inventory **ONCE**, ignore the 504, and verify via the API — never retry in the UI.

6. Recovery When Syncs Have Piled Up / Hung (verified 2026-07-27)

`set datasource: 400` is a **RED HERRING** — it is **NOT** the failure. The next log line is `DATA_SOURCE_ALREADY_CONFIGURED`; the tool continues. **The real abort is failed guardrails** (`DomainSyncException: ... has failed guard r`).

ails). Fix those with [[RUNBOOK-VCF-Sync-Inventory-Guardrail-Remediation-2026-07-27]] — *that* is what unblocks the sync (only `ERROR` guardrails block; `WARNING` s don't).

1. **Check whether the hung processes are alive or orphaned** — as `vcf`, no sudo: `bash ps -ef | grep vcf_brownfield | grep -v grep` The collided runs **stay alive** stuck in a 404-poll loop (they burn ~seconds of CPU over 90+ min) — they will **NOT** age out on their own on any useful timescale.
2. **Kill the hung sync PIDs** (as root via `su -`; the `vcf` sudo allowlist won't cover `kill`). **Kill the child PIDs, NOT the parent** (`4094388` -style = the domainmanager spawner — leave it). Use `-9` if TERM is ignored: `bash kill -9 <pid1> <pid2> ... # the vcf_brownfield.py children only` Once dead, the tasks self-reap to **Failed**, and `IN_PROGRESS` drops.
3. **Check for stale locks — often there are NONE:** `bash curl -sk -H "Authorization: Bearer $TOK" https://sddc-manager.lab.local/v1/resource-locks # 2026-07-27: 0` If `0`, **skip the DB lock-cleanup entirely** (it wasn't needed). Only if locks are held with no live task, use [[RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup-2026-06-09]].
4. **Fix the guardrails** (the actual blocker) per the guardrail runbook, then **click ONE clean Sync Inventory** and ignore the 504.
5. **Verify the result via the API, not the UI** — the new run's `results_mgmt.json` → `"status": "PASS"`, and `/v1/hosts` shows the hosts `ASSIGNED`.

Confirmed resolution 2026-07-27: guardrails cleared (esxi01 Management tag) → Sync domain mgmt = **PASS** → **all 4 hosts ASSIGNED**. Fully self-service, no Broadcom support, no DB surgery.

7. If It Cannot Be Reconciled

The hosts **still function** (they run workloads and serve vSphere) even while SDDC shows `status=None` — the impact is on **VCF lifecycle operations**, not runtime. Without support, an acceptable interim is to **leave the desync and monitor**, prioritizing the [[RCA-vpxd-Failure-Cluster-Store-InvalidState-2026-07-27]] / vSphere 9.1 upgrade (which re-mints the whole state) as the durable reset.

8. Changing ESXi Root While VCF Doesn't Manage It

Because the credential store holds no ESXi root (\$2), change ESXi root **directly in the Host Client** — it's safe (vCenter manages hosts via `vpxuser`, not root). See [[RUNBOOK-VCF-ESXi-vCenter-Root-Password-Change-2026-07-27]].

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Revision History

Version	Date	Notes
1.0	2026-07-27	Initial runbook.
1.1	2026-07-27	Updated §6 from the confirmed 2026-07-27 recovery: the hung sync processes are LIVE (404-poll), not orphaned, and must be killed (child PIDs, not the domainmanager parent); 0 stale locks meant no DB cleanup; the datasource-400 is a red herring, guardrails are the real blocker; confirmed resolution (sync PASS, all 4 hosts ASSIGNED).

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